



YASH THAKRE

AI Full Stack Engineer | C++, Python | GenAI & Agentic Systems

Roll No.: 23BTB0A33 | B.Tech (Minor in Management)

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EDUCATION

National Institute of Technology, Warangal

2023 – 2027

Bachelor of Technology (Minor in Management)

CGPA: 8.44

Lala Kailashpat Singhania School, Sausar

2022

Central Board of Secondary Education (CBSE Class XII)

Percentage: 87%

TECHNICAL SKILLS

Languages: C#, Python, C++, SQL (T-SQL, PostgreSQL, SQL Server), MATLAB, R

Full Stack & Backend: ASP.NET Core Web API, FastAPI, REST APIs, Microservices, Streamlit, OOP, System Design, DSA

GenAI & Agents: Retrieval-Augmented Generation (RAG), LangGraph, Semantic Kernel SDK, MCP, Azure OpenAI, ChromaDB

Cloud & DevOps: Microsoft Azure (App Services, Functions), Docker, Kubernetes, Azure DevOps, Git, GitHub Actions, CI/CD

AI Tools & Data: GitHub Copilot (AI-Assisted Dev), PyTorch, TensorFlow, Scikit-learn, XGBoost, DuckDB, Pandas, NumPy

EXPERIENCE

Deccan AI Experts

Hyderabad, India

AI Model Evaluation Intern (Remote)

May 2026 – Present

- Evaluated outputs across **5+ LLMs** via Python pipelines and REST APIs, assessing accuracy, safety, and RLHF alignment.
- Participated in Agile sprint planning and code reviews; built adversarial test suites surfacing **200+ edge cases** for model sprints.

Indian Institute of Technology (IIT) Hyderabad

Hyderabad, India

Research Intern, Applied Deep Learning (Advisor: Prof. Renu John)

June 2026 – July 2026

- Trained and benchmarked **8+ DL architectures** on a **68 GB** dataset using **PyTorch** with mixed-precision training.
- Enhanced reconstruction fidelity (**PSNR 24 to 34 dB**, **SSIM 0.61 to 0.91**) while cutting computational training runtime by **40%**.

Indian Institute of Technology (IIT) Bombay

Mumbai, India

Research Intern, Computational Modelling (Advisor: Prof. Sandeep Kalva)

May 2025 – July 2025

- Developed acoustic signal reconstruction pipelines in **MATLAB** using time-reversal backpropagation and delay-and-sum beamforming.
- Parallelized execution via **MATLAB parfor**, reducing scan runtime from 2 hours to <5 minutes (**24x speedup**).

KEY PROJECTS

StateFlow – Self-Correcting Agentic RAG Engine | LangGraph, Azure OpenAI, PostgreSQL, Docker GitHub

- Architected an end-to-end **Retrieval-Augmented Generation (RAG)** system using an 8-node cyclic **LangGraph** agent (CRAG/Self-RAG).
- Implemented hybrid dense/sparse retrieval (ChromaDB + BM25) via **REST APIs**, boosting **Faithfulness by 42%** and **Recall by 49%**.
- Containerized services with **Docker**, persisting state checkpoints in **PostgreSQL** with end-to-end **LangSmith** observability.

PrivAgent – Multi-Agent LLM Automation System | LangGraph, Semantic Kernel, MCP, Azure GitHub

- Designed **AI-powered applications** using multi-agent orchestration delegating tasks across Research, Web, and Code execution agents.
- Integrated **Semantic Kernel SDK** with fallback routing between local and **Azure OpenAI LLMs**, cutting inference costs by **90%**.
- Integrated enterprise tools and standard agent actions using the **Model Context Protocol (MCP)** for compliant workflows.

PulseStock – AI Demand Forecasting Platform | ASP.NET Core Web API, FastAPI, XGBoost, DuckDB GitHub

- Built automated **data pipelines** and zero-leakage ETL using **DuckDB window functions** and Pandera validation on 25k+ records.
- Trained an **XGBoost** model achieving **9.94% WAPE** (>90% accuracy) across 5 stores with automated 15% safety buffers.
- Shipped low-latency backend endpoints (<25ms) in **ASP.NET Core / FastAPI** with a Streamlit portal and **100% pytest CI coverage**.

ACHIEVEMENTS & LEADERSHIP

Research Publication Authored paper on deep learning reconstruction architectures for PAT imaging. 2026

Academic Excellence Secured rank in the **Top 15%** of the department batch at NIT Warangal. 2023 – Present

Competitive Programming Solved **460+ problems** on LeetCode covering Data Structures, Algorithms, and SQL tracks. 2023 – Present

Placement Coordinator, NITW Coordinated campus recruitment drives, company relations, and logistics. 2024 – Present